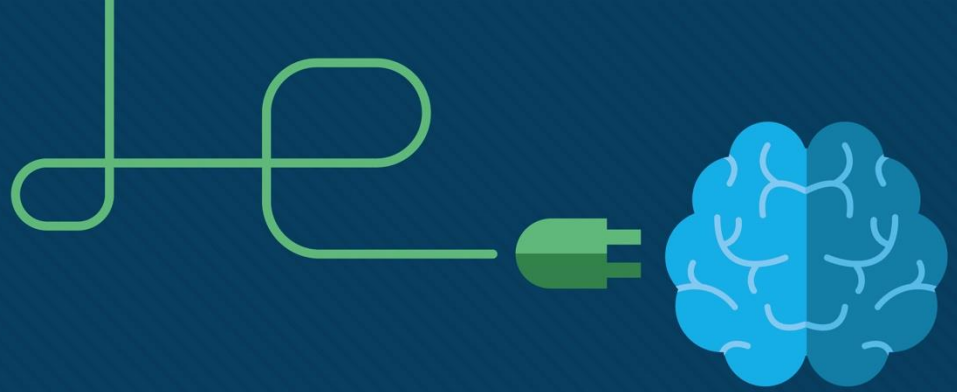

Module 9
Dynamic Routing

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Dynamic Routing

Module Objectives:

- Describe the characteristics and capabilities of dynamic routing protocols
- Describe the operations of the Routing Information Protocol
- Configure RIP in a network

Module References:

- CCNA R&S v6 Routing and Switching Essentials Module 3

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9.1 Introduction to Dynamic Routing

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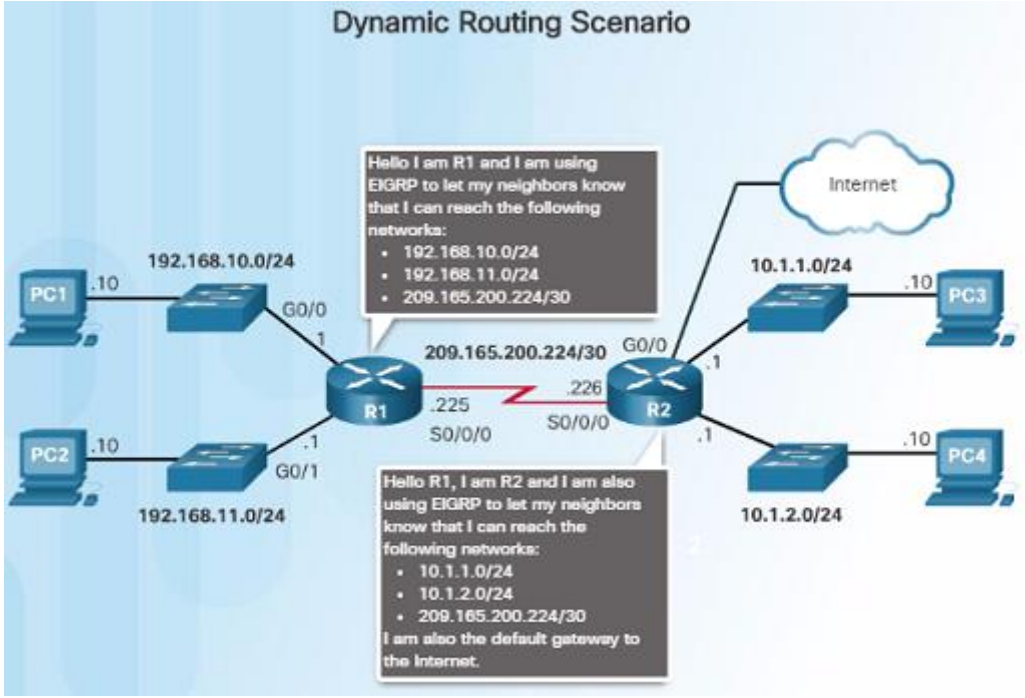
Dynamic Routing



- **Recall:** Dynamic routing is a networking technique enables routers to select paths according to the real-time logical network topology.
  - Routers automatically learn about remote networks from other routers
  - These networks and the best path to each are added to the routing table of the router.
- This technique is commonly implemented on in large networks to reduce the administrative overhead of configuring routes.

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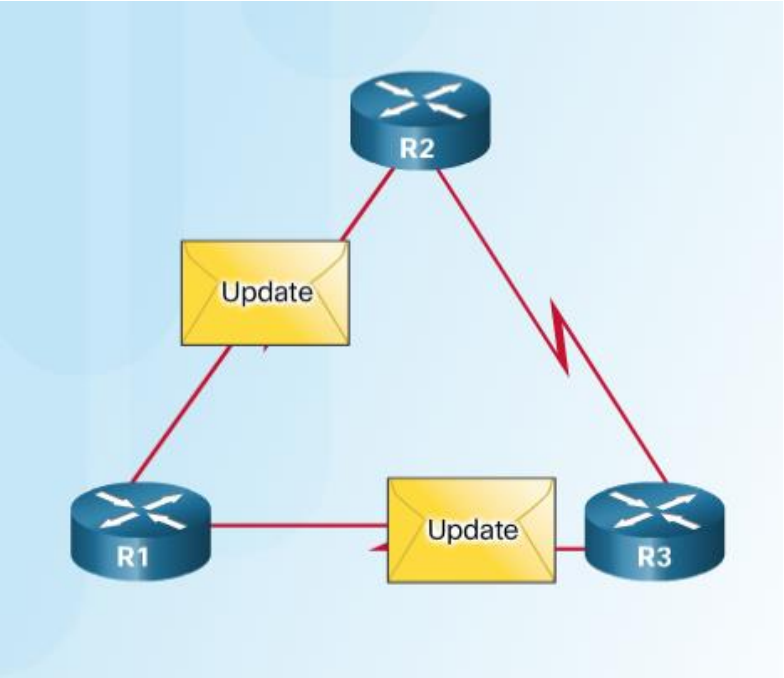
Dynamic Routing Protocols

- Dynamic routing protocols are used to facilitate the sharing of information about the reachability and status of remote networks between routers.
- The operations of dynamic routing protocols include:
  - Discovery of remote networks
  - Maintaining up-to-date routing information
  - Choosing the best path to destination networks
  - Ability to find a new best path if the current path is no longer available



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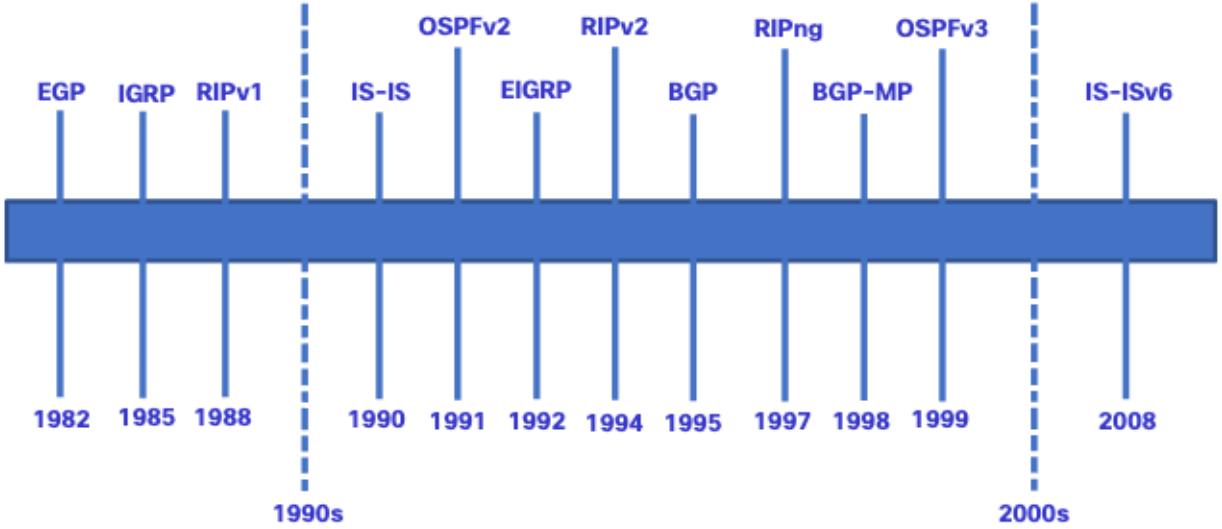
Dynamic Routing Protocol Evolution

Dynamic routing protocols have been used in networks since the late 1980s. One of the first routing protocols was RIP released in 1988, but some of the basic algorithms within the protocol were used on the Advanced Research Projects Agency Network (ARPANET) as early as 1969.

As networks evolved and became more complex, new routing protocols emerged.



Routing Protocol Classification

- Various routing protocols are available depending on network need.
- The table below lists some popular routing protocols used in IPv4 and IPv6 networks

	Interior Gateway Protocols				**Exterior Gateway Protocols**
	Distance Vector		Link-State		Path Vector
IPv4	RIPv2	EIGRP	OSPFv2	IS-IS	BGP-4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGP-MP

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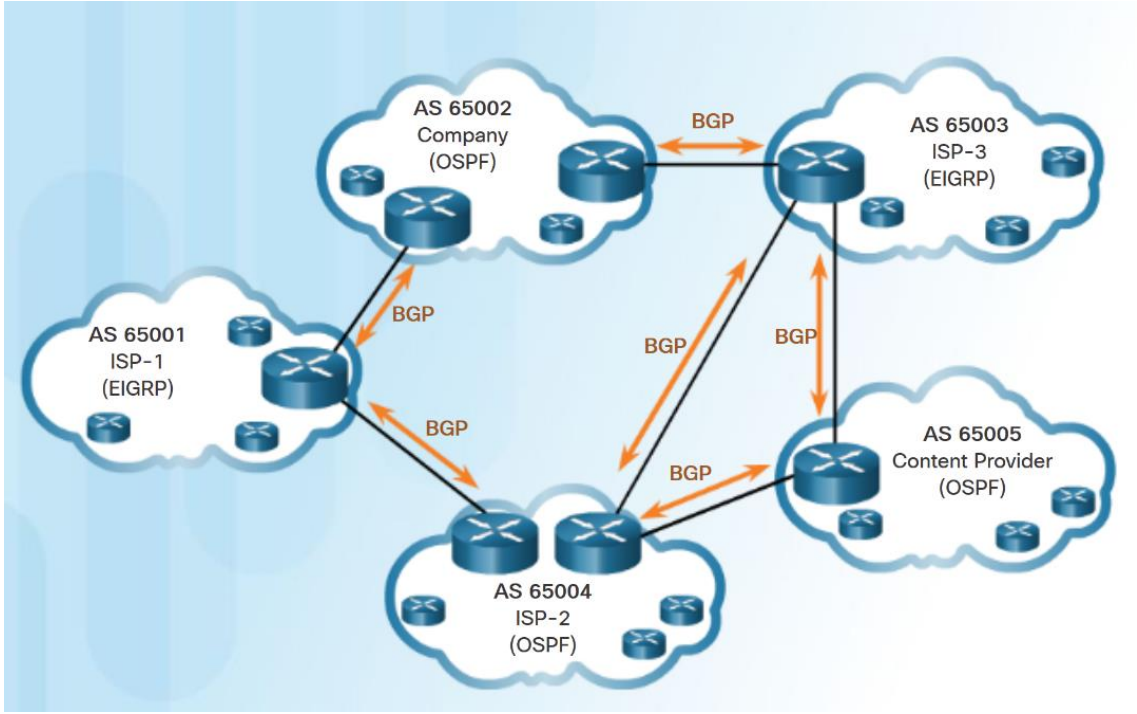
IGP and EGP Routing Protocols



- An Autonomous System (AS) is a network or set of networks under the administrative control of a single entity (e.g. a university, a company, etc)
- **Interior Gateway protocols (IGPs)** are used to exchange routing information between routers within an AS
- **Exterior Gateway Protocols (EGP)** are used for the exchange of routing information between routers from different AS

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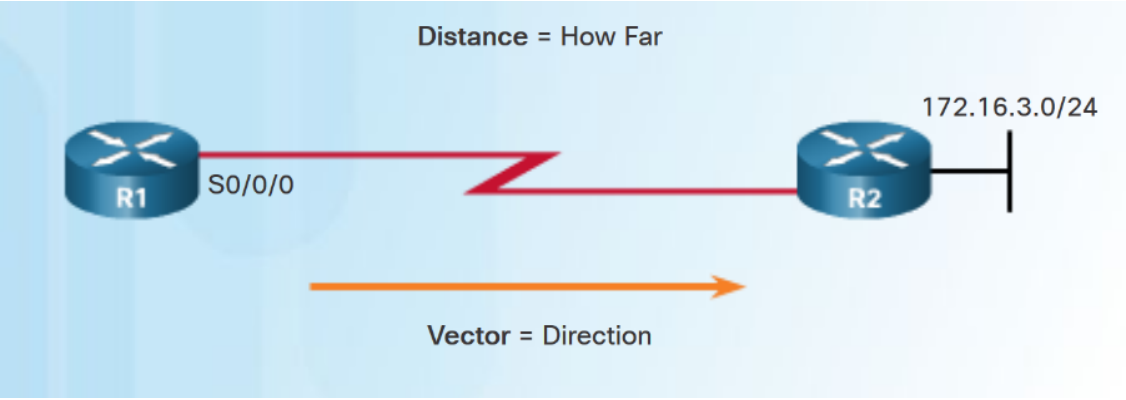
Distance Vector Routing Protocols

- Distance vector protocols are aware of reachable networks only by their distance and vector and do not know the complete path that packets take to their destination:
  - Distance - Identifies how far the destination network is based on a metric
  - Vector - Specifies the direction of the next-hop router or exit interface to reach the destination.
- Some examples: Routing Information Protocol (RIP), Enhance Interior Gateway Routing Protocol (EIGRP)



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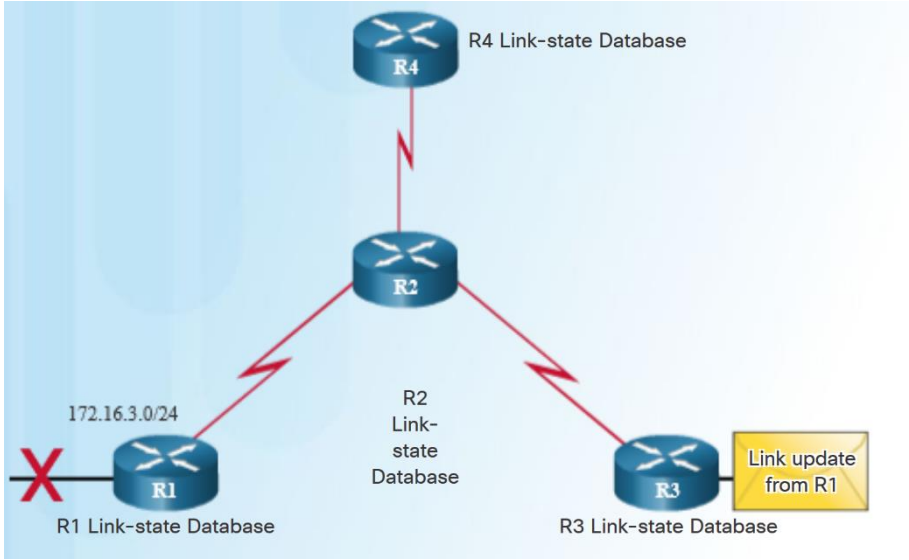
Link-State Routing Protocols



- Link state routing protocols learn about reachable networks by having a full view of the entire topology
- Router use link-state information received from other routers:
  - to create a topology map.
  - to select the best path to all destination networks in the topology.
- Some examples: Open Shortest Path First (OSPF), Intermediate System to Intermediate System (IS-IS)

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Dynamic Routing Protocol Components

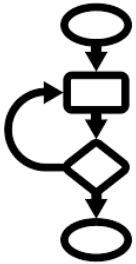
Main components of dynamic routing protocols include:

- **Data structures -** Routing protocols typically use tables or databases for its operations. This information is kept in RAM.

Net 1	…	…
Net 2	…	…

- **Routing protocol messages -** Routing protocols use various types of messages to discover neighboring routers, exchange routing information, and other tasks to learn and maintain accurate information about the network

- **Algorithm -** Routing protocols use algorithms for facilitating exchange of routing information and calculating the best path



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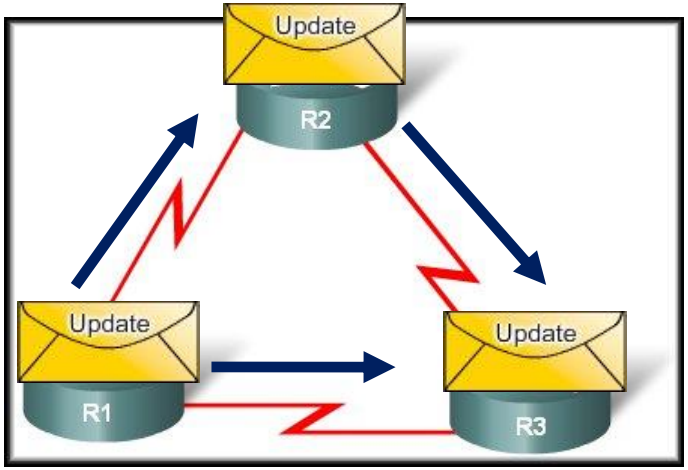
Dynamic Routing Protocol Operation

In general, the operations of a dynamic routing protocol can be described as follows:

1. The router sends and receives routing messages on its interfaces.
2. The router shares routing messages and routing information with other routers that are using the same routing protocol.
3. Routers exchange routing information to learn about remote networks.
4. When a router detects a topology change the routing protocol can advertise this change to other routers.



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Routing Protocol Metric

- Routing protocols use a metric to represent the distance to reach a network.
- Dynamic routing protocols typically use their own rules and metrics to build and update routing tables.

Routing Protocol	Metric
Routing Information Protocol (RIP)	• The metric is “hop count”. • Each router along a path adds a hop to the hop count.
Open Shortest Path First (OSPF)	• The metric is “cost” which is the based on the cumulative bandwidth from source to destination. • Faster links are assigned lower costs compared to slower (higher cost) links.
Enhanced Interior Gateway Routing Protocol (EIGRP)	• It calculates a metric based on the slowest bandwidth and delay values. • It could also include load and reliability into the metric calculation.

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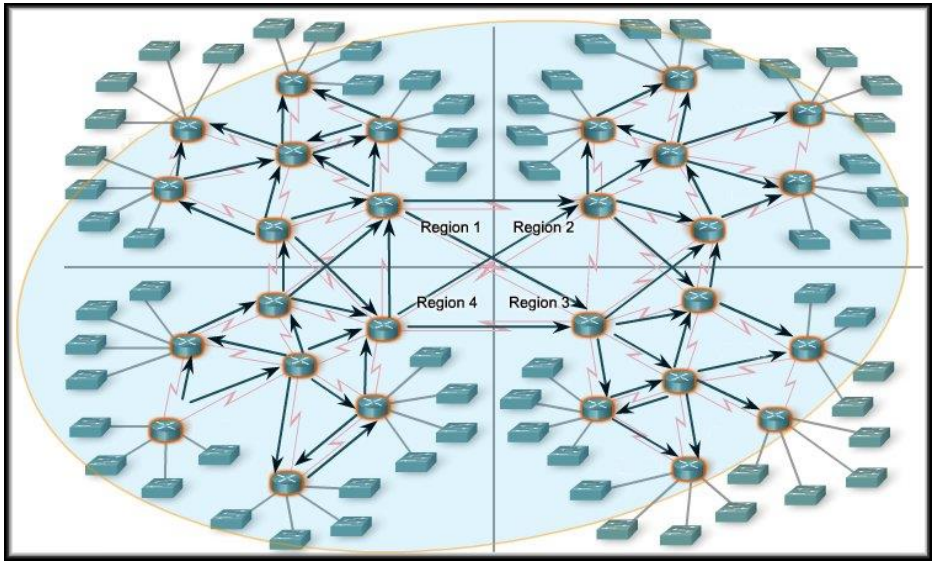
Achieving Convergence

- The network has *converged* when all routers have complete and accurate information about the network.
- The speed of convergence is an important characteristic of a network because it affects how quickly routers adjust to topology changes.
  - Affected by the size of the network
  - A network is not completely operable until it has converged.
  - Routing protocols with shorter convergence times are preferred.



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9.2 Routing Information Protocol

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Routing Information Protocol

- Is a distance vector routing protocol that uses hop count as its metric
- Uses the Bellman-Ford algorithm as its routing algorithm
- Limited to 15 hops maximum – A network more than 15 hops away is considered unreachable.
- Periodic routing updates - Entire routing table is broadcast every 30 seconds to neighbors.
- Assigned an AD of 120 on Cisco routers
- Available versions:
  - RIPv1 for IPv4 networks not using VLSM
  - RIPv2 for IPv4 networks using VLSM
  - RIPng for IPv6 networks

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How RIP Works – Starting Up

- After a router boots successfully and loads its saved configurations, it knows nothing about the network topology
- The router initially discovers its own directly connected networks and adds these to its routing table

Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0

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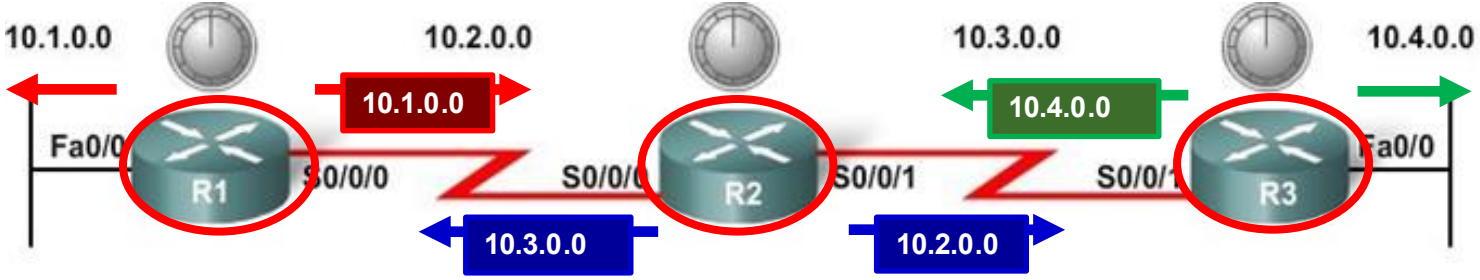
How RIP Works – Initial Exchange

- Routers begin sending routing updates containing their known networks to their neighbors
- Receiving neighbor increments the hop count of the received network and places the entry in the routing table



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0
10.2.0.0	S0/0/1	1

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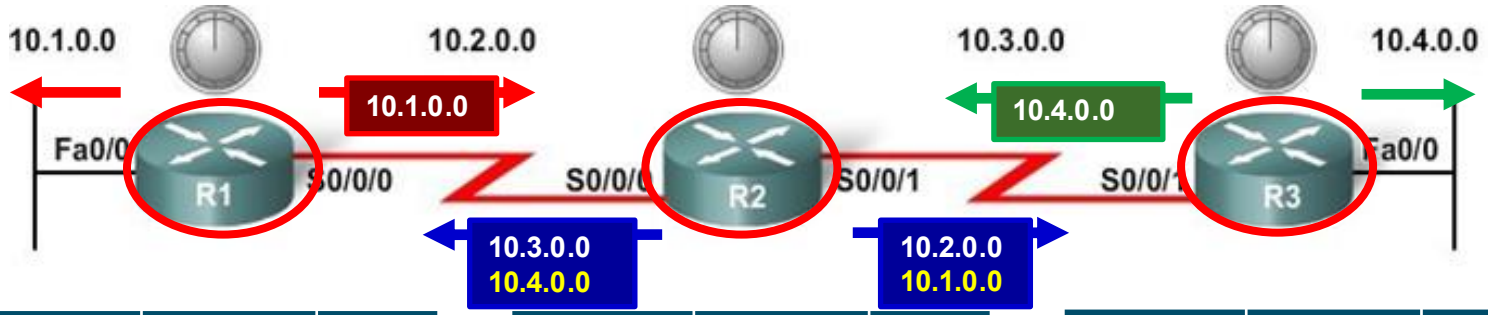
How RIP Works – Periodic Exchange

- The routers repeat the exchange at regular intervals (30 seconds), passing their respective route information to neighbors, including those that they learned from other routers.



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

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How RIP Works – Convergence and Maintenance

- Once all routers have the complete set of networks in their respective tables, the network has then converged, but routers continue to send periodic updates to keep tables up-to-date.
- Updates that do not contain new information are just ignored by the receiving router

Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

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Routing Loop Prevention

- Recall: A routing loop is a condition in which a packet is continuously transmitted within a series of routers without ever reaching its destination
- Routing loops can result in excess use of bandwidth and degraded network convergence
- In dynamic routing, loops may be caused by slow convergence and incorrect management of routing updates:

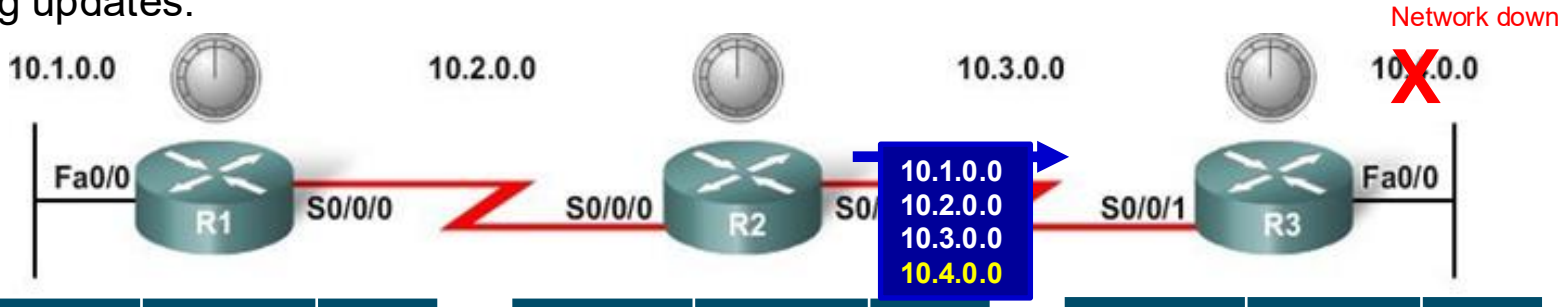
Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
~~10.4.0.0~~	~~F0/0~~	~~0~~
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

Routing Loop Prevention

- Recall: A routing loop is a condition in which a packet is continuously transmitted within a series of routers without ever reaching its destination
- Routing loops can result in excess use of bandwidth and degraded network convergence
- In dynamic routing, loops may be caused by slow convergence and incorrect management of routing updates:



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	S0/0/1	2
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

Routing Loop!

Routing Loop Prevention

- To prevent routing loops, RIP uses the following mechanisms:
  - Split horizon rule
  - Route poisoning
  - Hold-down timer

Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

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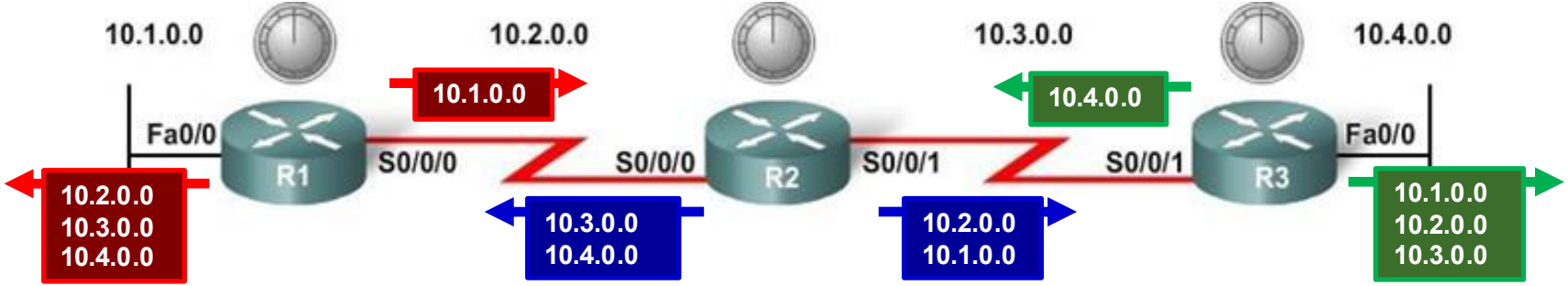
Routing Loop Prevention – Split Horizon

- The *split horizon rule* prevents routers from advertising a route out the same interface where it was received



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

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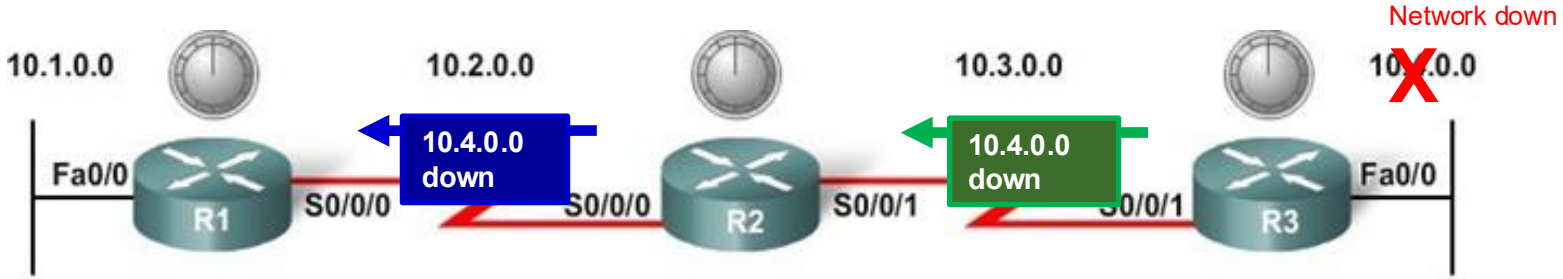
Routing Loop Prevention – Route Poisoning

- In route poisoning, a router that detects an unreachable network must deliberately advertise it as ‘down’ to its neighbor by tagging it with an unreachable metric (16)



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	16

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	16

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
10.4.0.0	F0/0	16
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

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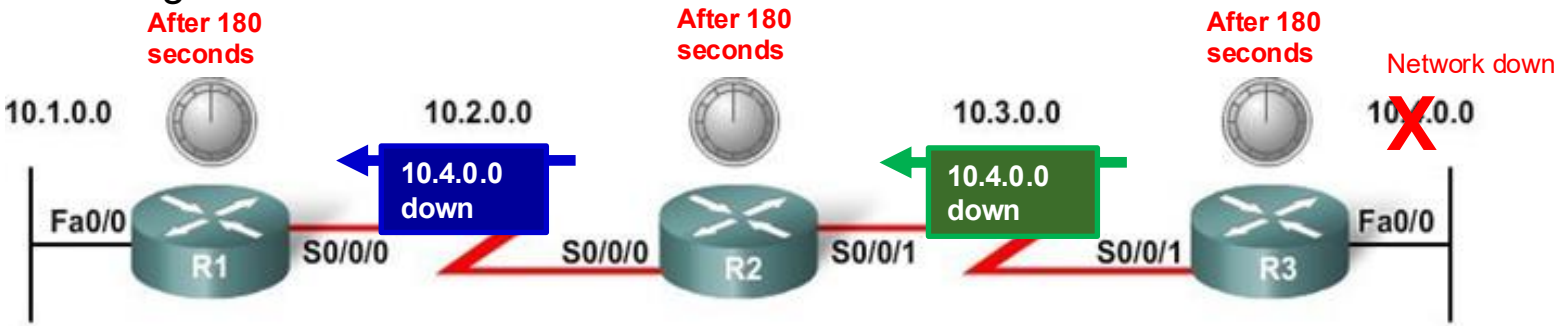
Routing Loop Prevention – Hold-down Timer

- Hold-down timers prevent a router from accepting further updates to a route for a specified period of time (180 seconds default) after it is advertised as unreachable (down)
- When the timer expires and a route is still being advertised as unreachable, it is removed from the routing table



Network	Interface	Hop Count
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
~~10.4.0.0~~	~~S0/0/0~~	~~16~~

Network	Interface	Hop Count
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
~~10.4.0.0~~	~~S0/0/1~~	~~16~~

Network	Interface	Hop Count
10.3.0.0	S0/0/1	0
~~10.4.0.0~~	~~F0/0~~	~~16~~
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

9.3 RIP Configuration for IPv4

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RIP Configuration Procedure

Task	IOS Command
1. Enter RIP configuration mode	**Router(config)# router rip**
2. Set RIP version (case-to-case basis)	**Router(config-router)# version *1|2***
3. Identify directly connected networks to advertise and participate in routing updates	**Router(config-router)# network** *net_address*
4. Select passive interfaces (optional but recommended)	**Router(config-router)# passive-interface** *interface_id*
5. Advertise a static default route (case-to-case basis)	**Router(config-router)# default-information originate**
6. Disable automatic route summarization (case-to-case basis)	**Router(config-router)# no auto-summary**

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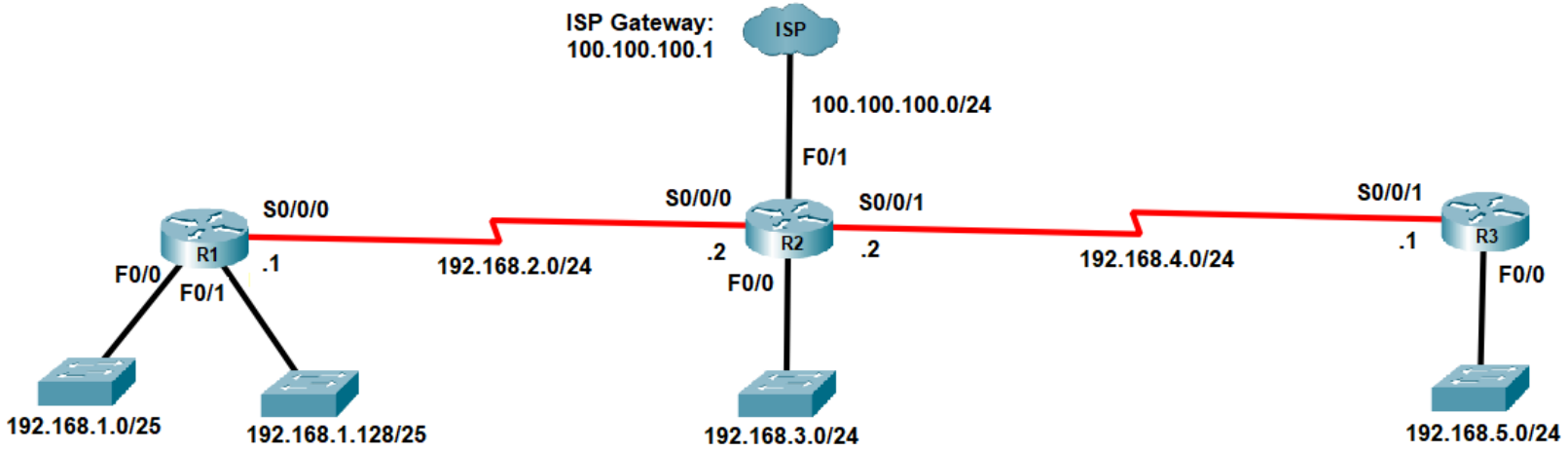
Router RIP Configuration Mode

- The ‘router’ command on global config mode is used to enable a routing protocol
- Default RIP version is version 1. Enable version 2 if the network uses different subnet masks




```
R1(config)#router rip
R1(config-router)#version 2
```

```
R2(config)#router rip
R2(config-router)#version 2
```

```
R3(config)#router rip
R3(config-router)#version 2
```

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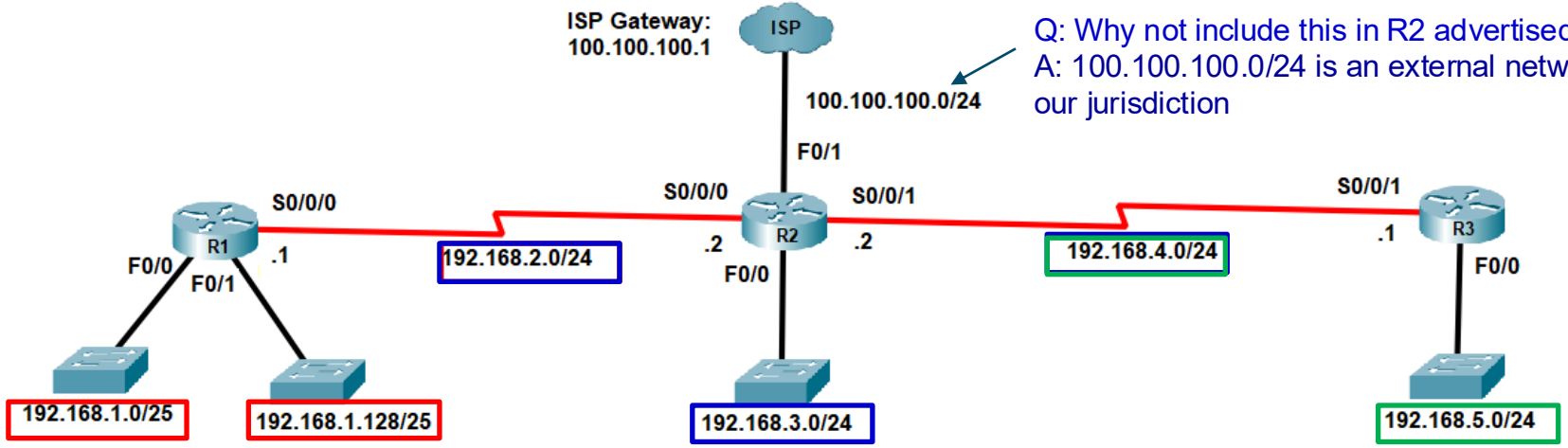
Advertising Networks

- The ‘network’ command enables RIP on all interfaces of a router that belong to the specified network and also directs the router to advertise known networks to neighbors connected on these interfaces



Q: Why not include this in R2 advertised networks?
A: 100.100.100.0/24 is an external network outside our jurisdiction

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.2.0
```

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 192.168.3.0
R2(config-router)#network 192.168.4.0
```

```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.4.0
R3(config-router)#network 192.168.5.0
```

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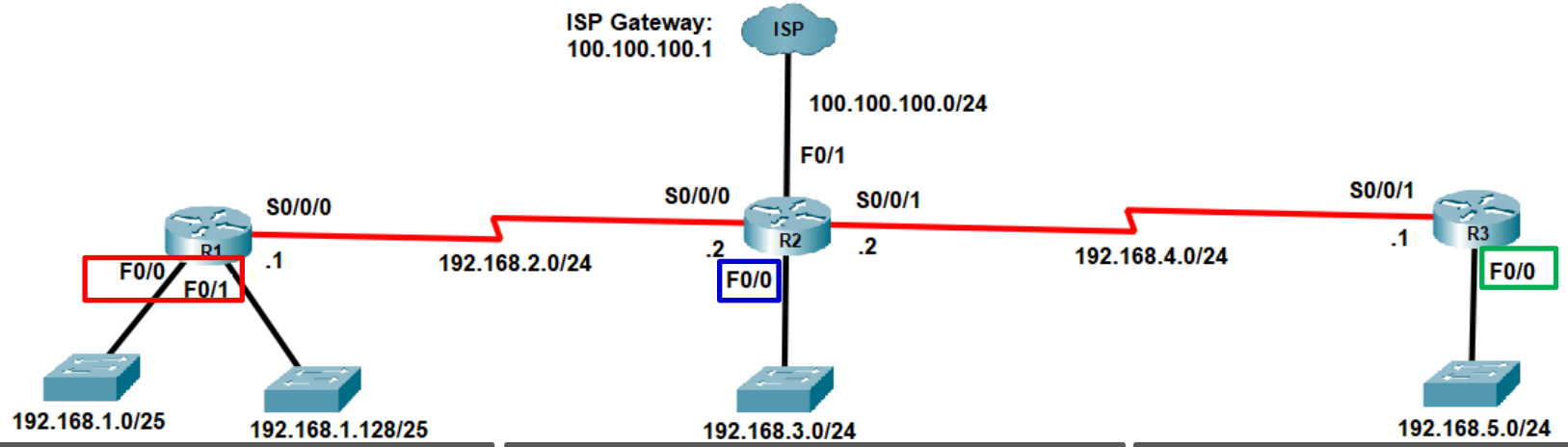
Setting Passive Interfaces

- RIP updates are forwarded out all RIP-enabled interfaces by default but actually only need to be sent out interfaces that are connected to other RIP-enabled routers.
- Sending RIP updates to user LANs wastes bandwidth, wastes resources, and is a security risk.
- To disable sending updates out an interface, set it to passive



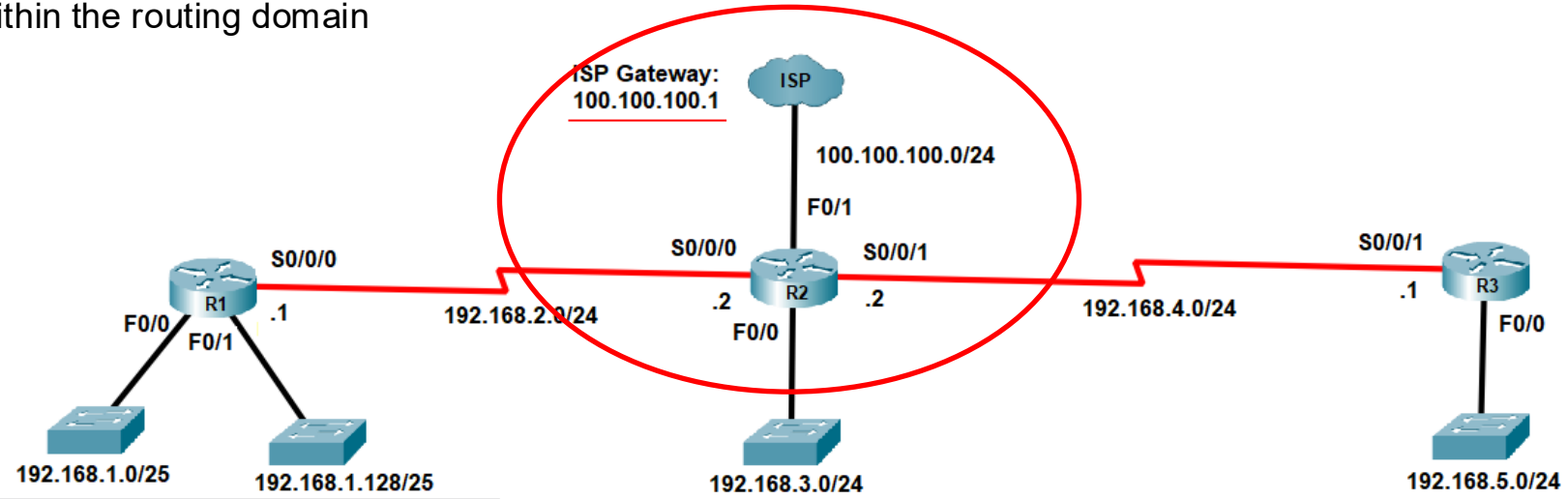

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.2.0
R1(config-router)#passive-interface f0/0
R1(config-router)#passive-interface f0/1
```

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 192.168.3.0
R2(config-router)#network 192.168.4.0
R2(config-router)#passive-interface f0/0
```

```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.4.0
R3(config-router)#network 192.168.5.0
R3(config-router)#passive-interface f0/0
```

Propagating a Default Route

- Edge routers connected to an external network can use RIP to pass default route information to other routers within the routing domain



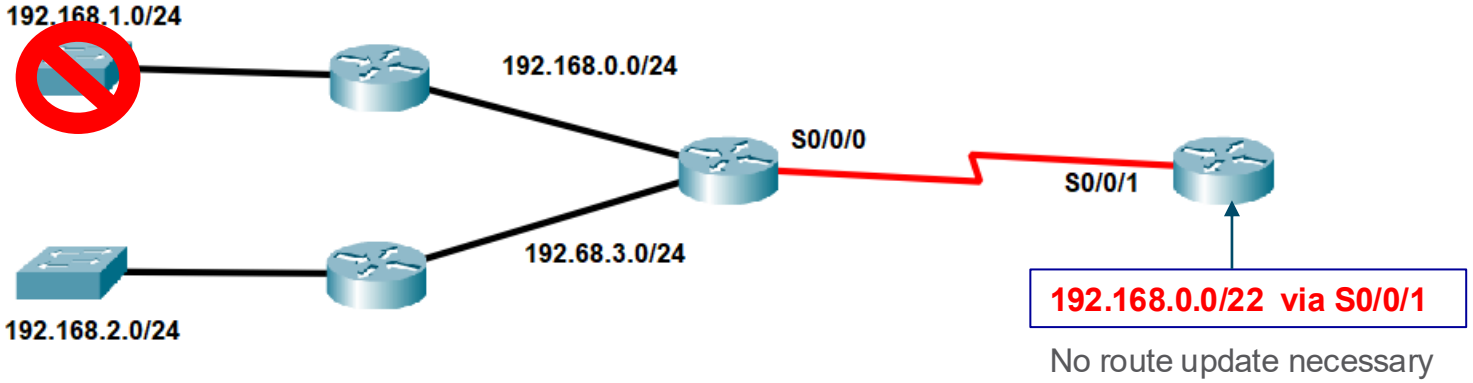

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.2.0
R1(config-router)#passive-interface f0/0
R1(config-router)#passive-interface f0/1
```

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 192.168.3.0
R2(config-router)#network 192.168.4.0
R2(config-router)#passive-interface f0/0
R2(config-router)#default-information
   originate
R2(config-router)#exit
R2(config)#ip route 0.0.0.0 0.0.0.0
   100.100.100.1
```

```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.4.0
R3(config-router)#network 192.168.5.0
R3(config-router)#passive-interface f0/0
```

Automatic Summarization

- Route summarization is commonly supported by routing protocols.
- When used together with a routing protocol, its benefits are:
  - Smaller routing tables thereby consuming less memory and allowing faster lookups
  - Smaller routing updates thereby consuming less network bandwidth
  - Less impact by topology changes. The summary network route does not need to be updated even if one of its subnets fails



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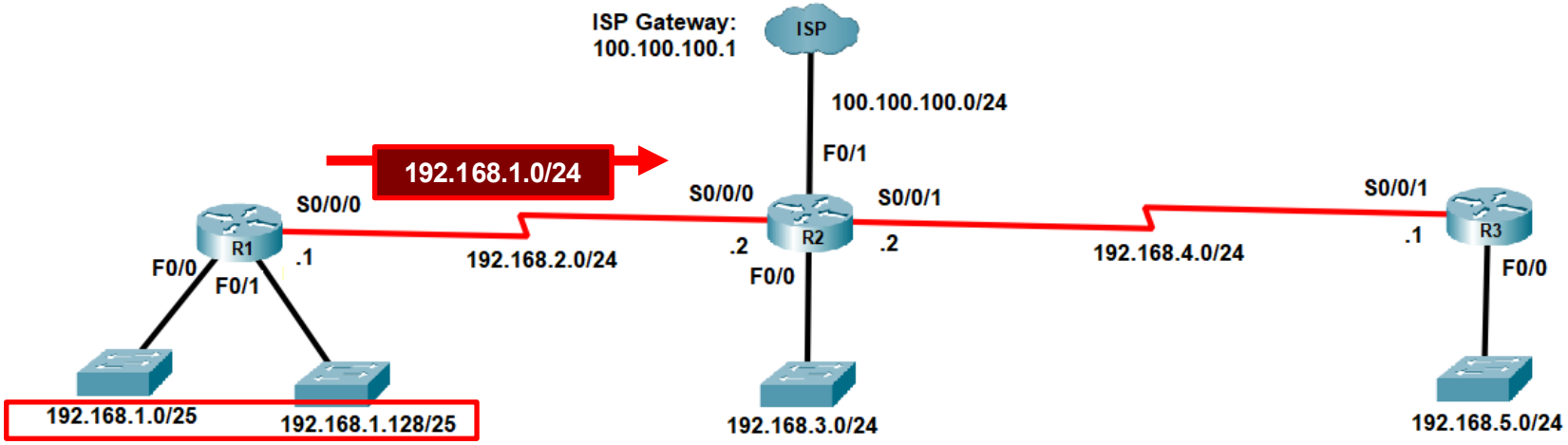
Automatic Summarization

- By default, RIP automatically summarizes networks into their classful boundaries according to their starting octet value when advertising routes to neighbors

Class A	1.0.0.0 – 127.0.0.0	/8
Class B	128.0.0.0 – 191.0.0.0	/16
Class C	192.0.0.0 – 223.0.0.0	/24



Subnets of class C network 192.168.1.0/24

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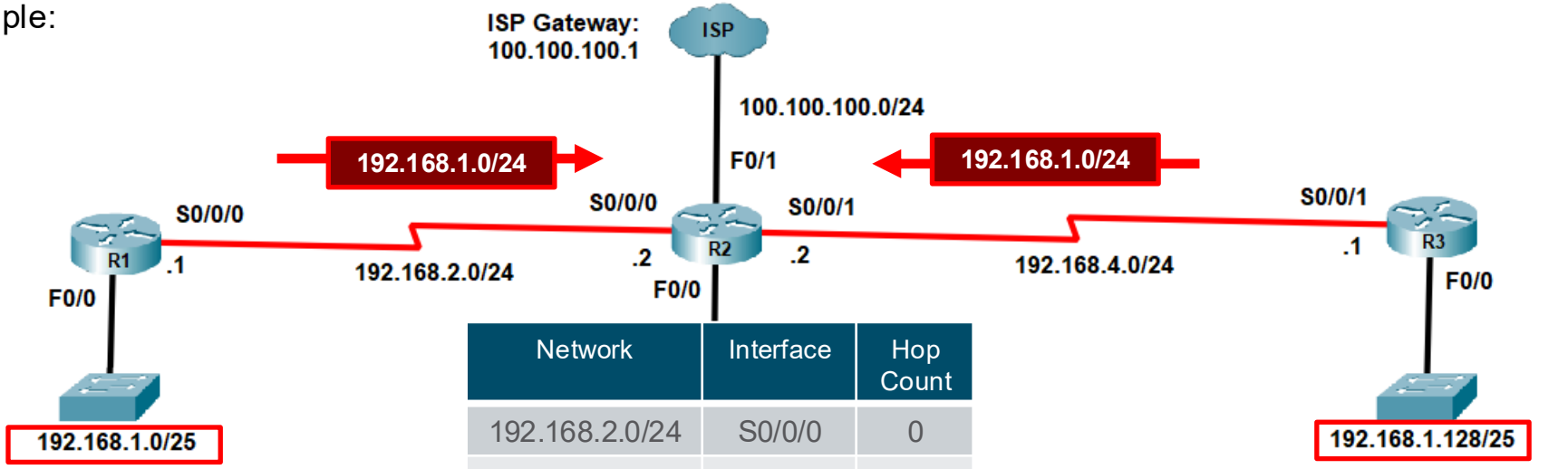
Automatic Summarization

- Automatic summarization should be disabled when the network topology uses discontiguous subnet allocation that can lead to incorrect routes advertised by routers
- Example:



Subnet of class C network
192.168.1.0/24

Subnet of class C network
192.168.1.0/24

Network	Interface	Hop Count
192.168.2.0/24	S0/0/0	0
192.168.3.0/24	F0/0	0
192.168.4.0/24	S0/0/1	0
192.168.1.0/24	S0/0/0	1
	S0/0/1	1

Equal cost load-balancing
(but should not be doing this)

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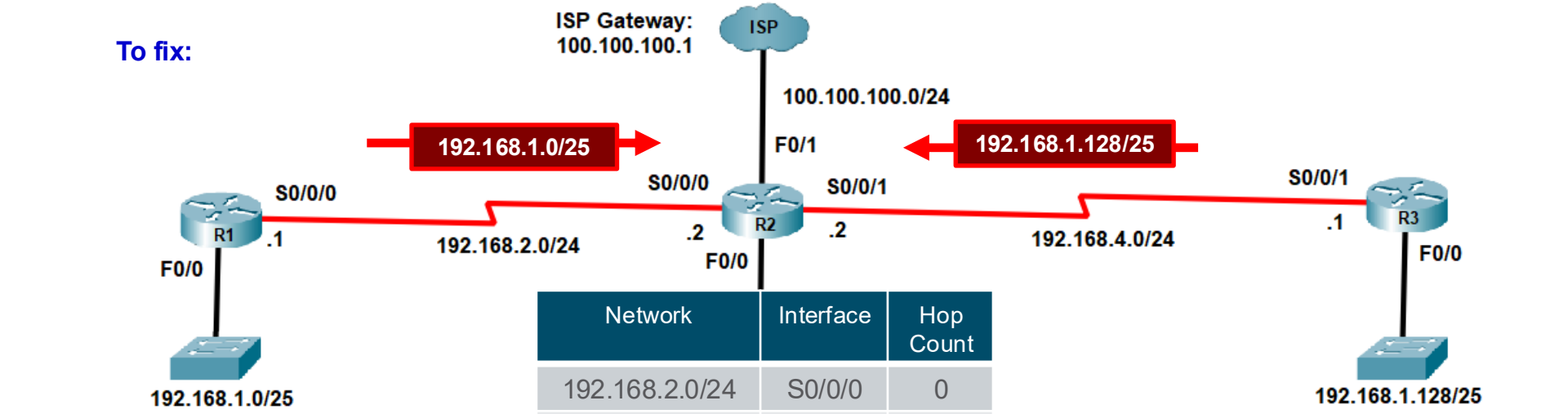
Automatic Summarization

- Automatic summarization should be disabled when the network topology uses discontiguous subnet allocation that can lead to incorrect routes advertised by routers

To fix:




```
R1(config)#router rip
R1(config-router)#no auto-summary
```

```
R3(config)#router rip
R3(config-router)#no auto-summary
```

Network	Interface	Hop Count
192.168.2.0/24	S0/0/0	0
192.168.3.0/24	F0/0	0
192.168.4.0/24	S0/0/1	0
192.168.1.0/25	S0/0/0	1
192.168.1.128/25	S0/0/1	1

Routes now correct

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Verify RIP Routing

show ip protocols – displays IPv4 routing protocols and operational settings of each protocol configured on the router.

```
R1#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 10 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  Serial0/0/0           2     2
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.1.0
  192.168.2.0
Passive Interface(s):
  FastEthernet0/0
  FastEthernet0/1
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.2.2          120      00:00:03
Distance: (default is 120)
```

Verify RIP Routing

show ip route – displays RIP (R) routes installed in the routing table and any default routes learned using RIP updates (R*).

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.2.2 to network 0.0.0.0

     192.168.1.0/25 is subnetted, 2 subnets
C       192.168.1.0 is directly connected, FastEthernet0/0
C       192.168.1.128 is directly connected, FastEthernet0/1
C    192.168.2.0/24 is directly connected, Serial0/0/0
R    192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:04, Serial0/0/0
R    192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:04, Serial0/0/0
R    192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:04, Serial0/0/0
R*   0.0.0.0/0 [120/1] via 192.168.2.2, 00:00:04, Serial0/0/0
```

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Questions?

What did you learn in this module?

- Dynamic routing enables routers to automatically share routes and adjust to topology changes by sharing route information using a routing protocol.
- Routing protocols are composed of data structures used to store route information, routing messages used for communication with other routers, and algorithms used for controlling protocol operations and calculations
- Routing protocols may be classified as
  - Interior vs exterior gateway protocols based on where they are configured
  - Distance vector vs link state based on the extent of their knowledge of the network topology
- Each routing protocol employs its own metric to represent a router's distance to a destination network
- RIP is a distance vector routing protocol where routers use hop count as the metric and send periodic updates to send routing table content to neighbors

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What did you learn in this module?

- To prevent routing loops from occurring, RIP uses the split horizon rule, route poisoning and hold-down timers
- To configure RIP on an IPv4 network
  - Enable RIP routing and select the version (v1 or v2) to use
  - Identify directly connected networks to advertise to neighbors
  - Set interfaces connecting to user LANs as passive
  - Propagate static default routes on an edge router
  - Disable auto-summarization on discontiguous LAN topologies

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